Last class we looked at the power transmitted by waves:

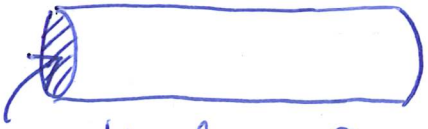
transverse:




$$\langle P \rangle = \frac{1}{2} Z \omega^2 A^2$$

where $Z = \sqrt{T\mu}$ mechanical impedance

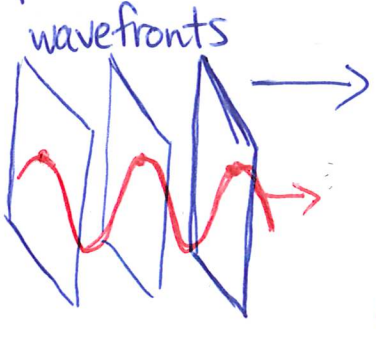
longitudinal:




$$\langle P \rangle = \frac{1}{2} Z S \omega^2 A^2$$

where $Z = \sqrt{Y\rho}$ acoustic impedance

plane waves in fluid:




$$\langle I \rangle = \frac{1}{2} Z \omega^2 A^2$$

intensity

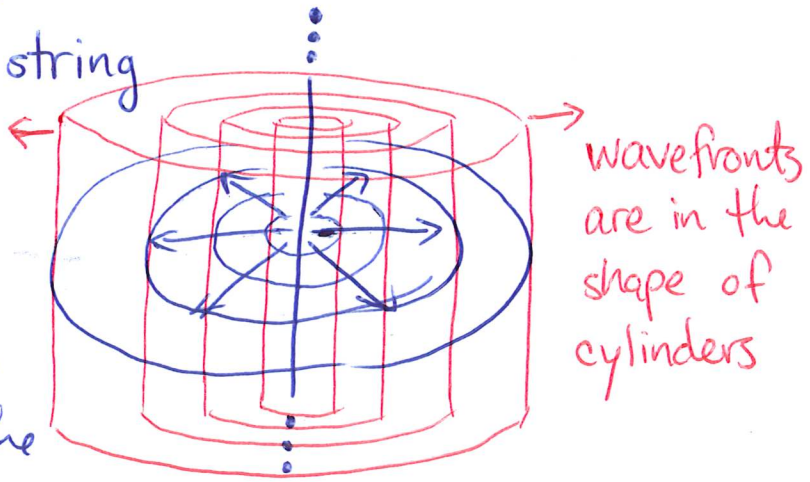
where $Z = \sqrt{B\rho}$ acoustic impedance

for fluids, rather than the <u>total</u> power carried by the wave, it is more useful to quantify the power per unit area, known as the <u>intensity</u>. For plane waves, the intensity remains constant as the wave travels because the energy is not spreading out and we have assumed ideal fluids with no attenuation (energy lost to heat).

Let's now look at more realistic situations of sound waves travelling in 2D and 3D.

2D: long vibrating string

due to symmetry, the wavefronts will travel radially outward from the string producing the sound.



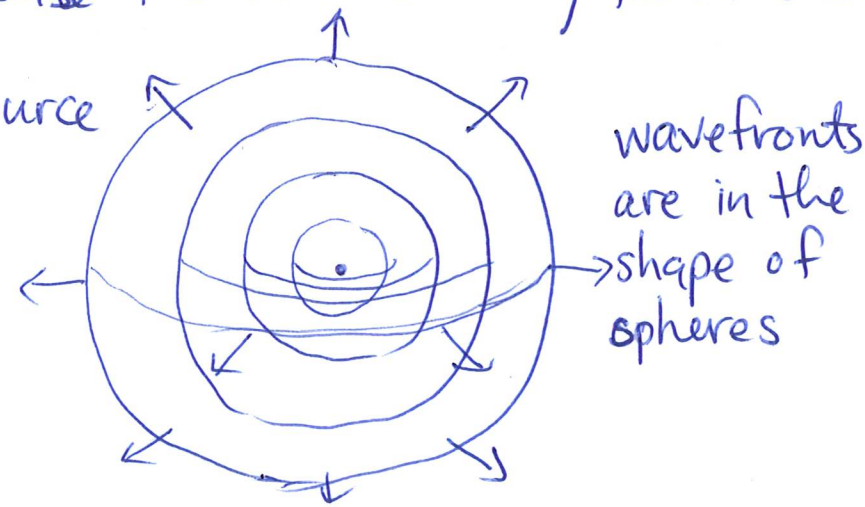

The power (total energy per unit time) must be constant and we know that in a fluid like air

$$\langle I \rangle = \frac{\langle P \rangle}{S} \qquad \text{but } S = 2\pi r L$$

so $\boxed{\langle I \rangle = \frac{C}{r}}$ where $C$ = constant

The intensity of the sound decreases like $\frac{1}{r}$, so the sound is less intense the further away from the string.

3D: point source



Using the same argument as the 2D case:

$$\langle I \rangle = \frac{\langle P \rangle}{S} \qquad \text{but} \quad S = 4\pi r^2$$

$$\text{so} \quad \boxed{\langle I \rangle = \frac{C}{r^2}} \qquad \text{where} \quad C = \text{constant}$$

When waves spread out from point sources, the intensity falls off as $\frac{1}{r^2}$.

Both of these cases are examples of <u>geometric spreading</u> and are direct consequences of conservation of energy.

In addition to this, real media are not ideal and as the waves propagate, energy is lost to heat (friction, random motion of particles, collisions, etc.) The rate at which intensity is lost by an acoustic wave is proportional to the intensity itself (bigger oscillations lead to quicker loss, smaller oscillations lead to slower loss)

$$\frac{dI}{dx} = -\alpha I \qquad \alpha = \text{"attenuation coefficient"}$$
units are $m^{-1}$.

the solution is $I(x) = I_u e^{-\alpha x}$

$I_u$ ↑ unattenuated intensity.

For plane waves: $I(x) = I_0 e^{-\alpha x}$

2D waves: $I(r) = \frac{C}{r} e^{-\alpha r}$

3D waves: $I(r) = \frac{C}{r^2} e^{-\alpha r}$

5 minute break

Example: A point source emits sound at 20W. What is the unattenuated intensity at 100m?

$I_u = \frac{P}{S}$ where $S = 4\pi r^2$ with $r = 100\text{m}$

$\Rightarrow I_u = \frac{20\text{W}}{4\pi(100\text{m})^2} = \boxed{1.59 \times 10^{-4}\ \text{W/m}^2}$

If the attenuation coefficient of air is $\alpha = 0.01\ \text{m}^{-1}$, what is the attenuated intensity at 100m?

$I(100\text{m}) = I_u e^{-\alpha r}$

$= (1.59 \times 10^{-4}\ \text{W/m}^2)\, e^{-(0.01\text{m}^{-1})(100\text{m})}$

$\boxed{I(100\text{m}) = 5.85 \times 10^{-5}\ \text{W/m}^2}$

The power of a sound wave transmitted through our eardrums is $\langle P \rangle_{ear} = \langle I \rangle S_{eardrum}$

which is why the intensity is a more useful quantity when talking about a source's sound level.

The decibel scale: based on how we perceive stimulus changes (Weber's Law)

the sound level $\beta$ of an acoustic wave is a logarithmic scale:

$$\beta = 10 \log\left(\frac{I}{I_0}\right)$$ ($\log$: base 10) where $I_0 = 10^{-12}$ W/m² is the threshold of hearing

When $I = I_0$ (at the threshold) $\log(1) = 0$

$\Rightarrow \beta(I_0) = 0$ dB

When $I = 10 I_0$: $\log(10) = 1 \Rightarrow \beta(10 I_0) = 10$ dB

$I = 100 I_0$: $\log(100) = \log(10^2) = 2 \Rightarrow \beta(100 I_0) = 20$ dB

an intensity 10 times larger is perceived as a sound that is 2 times louder.

$I = 10^n I_0$: $\log(10^n) = n \Rightarrow \beta(10^n I_0) = 10n$ dB

each time the intensity grows by an order of magnitude, the sound level increases by 10 $\Rightarrow$ logarithmic scale.

Example: A point source gives off sound at 50W in a medium with attenuation coefficient $\alpha = 0.035\ m^{-1}$. What is the sound level at a distance of 25m from the source?

What we need: $\beta(25m) = 10\log\left(\frac{I(25m)}{I_0}\right)$

First calculate $I(25m) = I_u e^{-\alpha r}$

$\Rightarrow$ need $I_u = \frac{P}{S(r)} = \frac{50W}{4\pi(25m)^2} = 6.37\times10^{-3}\ W/m^2$

therefore $I(25m) = (6.37\times10^{-3}\ W/m^2)\, e^{-(0.035m^{-1})(25m)}$

$= 2.66\times10^{-3}\ W/m^2$

Now we can find the sound level at 25m:

$$\beta(25m) = 10\log\left(\frac{2.66\times10^{-3}\ W/m^2}{10^{-12}\ W/m^2}\right)$$

$$\boxed{\beta(25m) = 84.3\ dB}$$

for comparison, a typical food blender at operating distance is $\sim$ 88 dB

* Sounds at 85 dB can lead to hearing loss when exposed for long time periods (hours). Nightclubs are typically at $\sim$106 dB $\Rightarrow$ use ear plugs!